

Pac-Man

User Documentation

Game States:

- **Title Screen:**

You will see this screen once you start up the game. You will notice a prompt at the bottom of the screen telling you to “press spacebar to play”. Follow this instruction to start the game.

- **Gameplay State:**

You will enter this state once you press spacebar to leave the title screen. The following objects will be visible to you:

1. The maze of the game will appear along with the four ghosts, the score circles and Pac-man.
2. There will be several powerups (read below to learn more about these).
3. On the top of the screen will be your current recorded score and on it's left will be your remaining lives in the form of two Pac-men initially corresponding to two lives.

Once you enter the gameplay state, a countdown of three seconds (visible on the lower half of the screen) will begin. Read the gameplay section of the documentation to learn what happens once the countdown elapses.

- **Game Over:**

You will head to this screen after either losing the game or winning it and you will get a background accordingly.

Gameplay:

- **Overview:**

Just like a standard Pac-man game, you will control Pac-man to eat as many score circles as possible while evading the four ghosts: Blinky, Pinky, Inky, Clyde. If you eat all the score circles you will win the game. If you get eaten by a ghost, you lose one life. Losing all lives results in a loss.

- **Ghosts:**

To win the game, you must avoid the ghosts at all costs. For simplicity, the ghosts have a random AI, although the ghosts are released at specific intervals from their initial position in the following order: Blinky, Pinky, Inky, Clyde.

- **Scoring:**

By eating each score circle, your Pac-Man gains 10 points while eating a ghost gets you 100 points. The number of points does not determine when you win the game – there is no score limit. Your objective is only to eat all the score circles. The score is there to record your performance and compare it to your last high score.

- **Power-Ups:**

There are several power-ups in the game granting different abilities. These do not respawn with the exception of teleporters, so it is recommended to use them wisely and only when needed.

- **Dynamic Maze Elements:**

You will notice that both the speed boost power-ups (bolt) are blocked by walls which appear or disappear every 5 seconds, so you need to plan your escape routes accordingly. Additionally, moving through one vertical edge of the window will move you towards the other edge.

Power-Up Effects:

-  Disables any wall collisions and allows Pac-man to phase through any walls. However, if the power ends while you're inside a wall, your Pac-Man will face instant death.
-  Halves Pac-man's speed.
-  Significantly increases Pac-man's speed.
-  Two teleporters that portal Pac-Man to the other teleporter's destination. Both teleporters spawn every 10 seconds after being used. Similarly, using one teleporter uses up both.

Controls:

Spacebar - used to start the game from the start screen

Up arrow key - moves upwards

Down arrow key - moves downwards

Left arrow key - moves left

Right arrow key - moves right